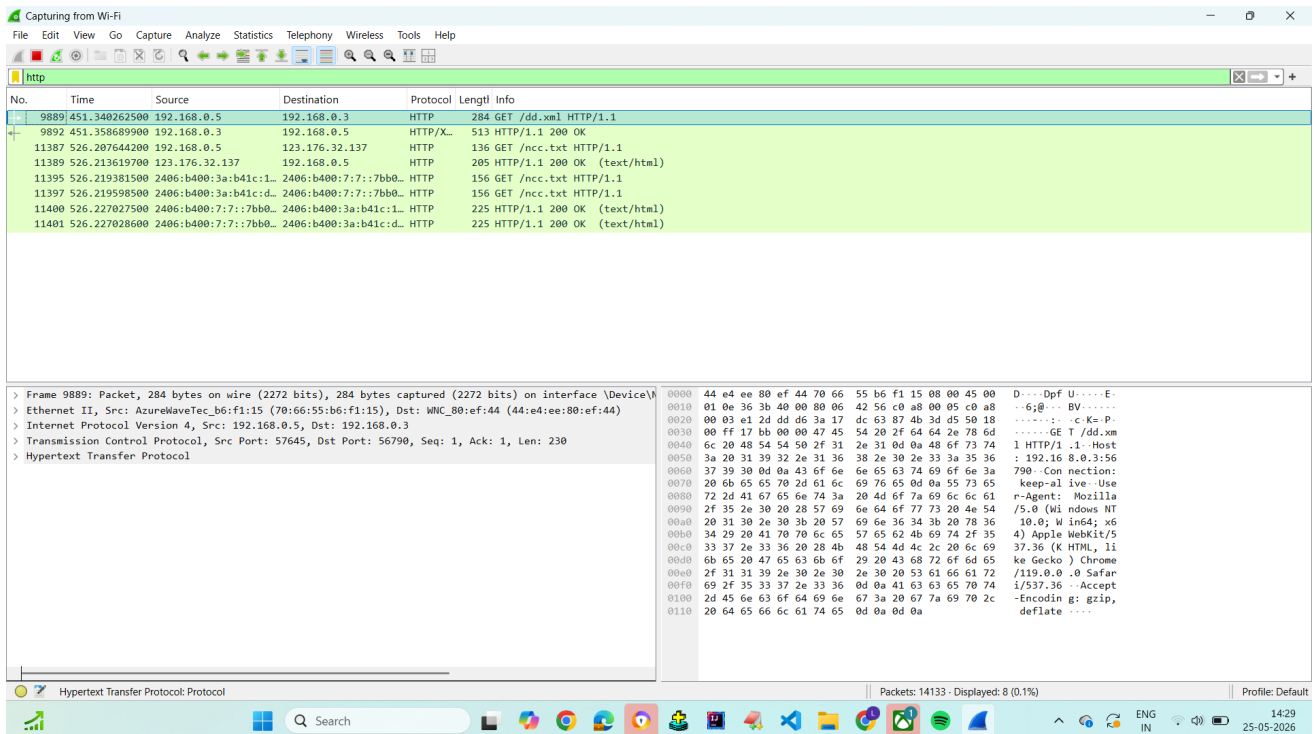


WIRESHARK NETWORK TRAFFIC ANALYSIS REPORT

This report demonstrates basic network traffic analysis using Wireshark. The captured packets include HTTP traffic, DNS queries, TCP three-way handshake packets, and TCP analysis flags such as retransmissions and duplicate acknowledgments.

1. HTTP Traffic Analysis

The HTTP filter was applied in Wireshark to capture web traffic packets. The screenshot shows HTTP GET requests and HTTP 200 OK responses between source and destination IP addresses. This demonstrates how browsers communicate with web servers using the HTTP protocol.



2. DNS Traffic Analysis

The DNS filter was used to capture Domain Name System queries and responses. The packets show DNS lookups for domain names such as google DNS and cloudflare DNS. DNS converts human-readable domain names into IP addresses.

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dns

No.	dns	Source	Destination	Protocol	Length	Info
1761	70.817955200	192.168.0.5	192.168.0.1	DNS	198	Standard query response 0x1051 A ncc.avast.com CNAME ncc.avast.com.edgesuite.net CNAME a1488.dscd.akamai.net A 123.176.32.137 A 123...
1762	70.818557800	192.168.0.5	192.168.0.1	DNS	93	Standard query 0x0fe6 A dns.google OPT
1765	70.908732600	192.168.0.1	192.168.0.5	DNS	93	Standard query 0x6d71 AAAA dns.google OPT
1766	70.908733400	192.168.0.1	192.168.0.5	DNS	113	Standard query response 0x0fe6 A dns.google A 8.8.4.4 A 8.8.8.8 OPT
1835	71.435501600	192.168.0.5	192.168.0.1	DNS	137	Standard query response 0x6d71 AAAA dns.google AAAA 2001:4860:4860::8844 AAAA 2001:4860:4860::8888 OPT
1836	71.435597500	192.168.0.5	192.168.0.1	DNS	101	Standard query 0x4e5a A cloudflare-dns.com OPT
1839	71.522976900	192.168.0.1	192.168.0.5	DNS	101	Standard query 0xe376 AAAA cloudflare-dns.com OPT
1840	71.522977300	192.168.0.1	192.168.0.5	DNS	121	Standard query response 0x4e5a A cloudflare-dns.com A 104.16.248.249 A 104.16.249.249 OPT
1929	73.393140700	fe80::aef5:6532:dd9...	fe80::eef:15ff:fed...	DNS	145	Standard query response 0x376 AAAA cloudflare-dns.com AAAA 2606:4700::6810:f9f9 AAAA 2606:4700::6810:f8f9 OPT
1930	73.393243600	fe80::aef5:6532:dd9...	fe80::eef:15ff:fed...	DNS	104	Standard query 0xb65f A ipv6.msftconnecttest.com
1931	73.398045800	fe80::eef:15ff:fed...	fe80::aef5:6532:dd9...	DNS	249	Standard query response 0xb65f A ipv6.msftconnecttest.com CNAME ipv6.msftconnecttest.com.edgesuite.net CNAME a1968.i6gl.akamai.net ...
1932	73.398046900	fe80::eef:15ff:fed...	fe80::aef5:6532:dd9...	DNS	244	Standard query response 0xb65f AAAA ipv6.msftconnecttest.com CNAME ipv6.msftconnecttest.com.edgesuite.net CNAME a1968.i6gl.akamai.net ...
1952	74.674403400	fe80::aef5:6532:dd9...	fe80::eef:15ff:fed...	DNS	112	Standard query 0xb6cd AAAA mobile.events.data.microsoft.com
1953	74.674450400	fe80::aef5:6532:dd9...	fe80::eef:15ff:fed...	DNS	112	Standard query 0x76b4 A mobile.events.data.microsoft.com
1954	74.679672100	fe80::eef:15ff:fed...	fe80::aef5:6532:dd9...	DNS	277	Standard query response 0xb6cd AAAA mobile.events.data.microsoft.com CNAME mobile.events.data.trafficmanager.net CNAME onedscoprd...
1955	74.679673400	fe80::eef:15ff:fed...	fe80::aef5:6532:dd9...	DNS	232	Standard query response 0x76b4 A mobile.events.data.microsoft.com CNAME mobile.events.data.trafficmanager.net CNAME onedscoprd...

> Frame 88: Packet, 91 bytes on wire (728 bits), 91 bytes captured (728 bits) on interface \Device\NPF...
> Ethernet II, Src: AzureWaveTec_b6:f1:15 (70:66:55:b6:f1:15), Dst: TPLink_fd:83:58 (0c:ef:15:fd:83:58)
> Internet Protocol Version 6, Src: fe80::aef5:6532:dd95:67fe, Dst: fe80::eef:15ff:fed:8358
> User Datagram Protocol, Src Port: 59634, Dst Port: 53
> Domain Name System (query)

Domain Name System: Protocol

Packets: 1986 - Displayed: 120 (6.0%)

Profile: Default

41°C Partly sunny

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3. TCP Three-Way Handshake Analysis

The TCP SYN filter was applied to observe TCP connection establishment. The highlighted packets show SYN and SYN-ACK packets exchanged between systems. This process is known as the TCP Three-Way Handshake and is used to establish reliable communication. Important Packet Numbers: - Frame 24272 : SYN packet - Frame 24273 : SYN-ACK packet

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tcp.flags.syn == 1

No.	Time	Source	Destination	Protocol	Length	Info
24152	889.623006000	192.168.0.1	192.168.0.5	TCP	74	46204 → 135 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM TSval=9416902 TSecr=0 WS=32
24153	889.722308400	192.168.0.1	192.168.0.5	TCP	74	35812 → 139 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM TSval=9416912 TSecr=0 WS=32
24155	889.821587900	192.168.0.1	192.168.0.5	TCP	74	47092 → 445 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM TSval=9416922 TSecr=0 WS=32
24156	889.922735800	192.168.0.1	192.168.0.5	TCP	74	43844 → 1883 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM TSval=9416932 TSecr=0 WS=32
24164	890.023221800	192.168.0.1	192.168.0.5	TCP	74	52376 → 8883 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM TSval=9416942 TSecr=0 WS=32
24239	895.073437600	2406:b400:3a:b41c:1...	2603:1046:2080:90:4	TCP	86	21878 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1420 WS=256 SACK_PERM
24240	895.095139300	2603:1046:2080:90:4	2406:b400:3a:b41c:1...	TCP	86	443 → 21878 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 WS=256 SACK_PERM
24272	895.645888300	192.168.0.5	192.168.0.1	TCP	66	21879 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=4 SACK_PERM
24273	895.648980000	192.168.0.1	192.168.0.5	TCP	66	80 → 21879 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1460 SACK_PERM WS=32
24338	896.082040700	192.168.0.5	13.89.179.10	TCP	66	21880 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
24345	896.461274500	13.89.179.10	192.168.0.5	TCP	66	443 → 21880 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
24380	897.527402900	2406:b400:3a:b41c:1...	2404:6800:4007:838...	TCP	86	21881 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1420 WS=256 SACK_PERM
24381	897.546095500	2404:6800:4007:838...	2406:b400:3a:b41c:1...	TCP	86	80 → 21881 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1420 SACK_PERM WS=256
24402	898.090371300	2406:b400:3a:b41c:1...	2600:140f:9:38d::21	TCP	86	21882 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1420 WS=256 SACK_PERM
24403	898.094502800	2600:140f:9:38d::21	2406:b400:3a:b41c:1...	TCP	86	80 → 21882 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1420 SACK_PERM WS=128
24416	898.426789200	2406:b400:3a:b41c:1...	2600:14e1:18:6d::	TCP	86	21883 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1420 WS=256 SACK_PERM
24417	898.452114600	2600:14e1:18:6d::	2406:b400:3a:b41c:1...	TCP	86	80 → 21883 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1328 SACK_PERM WS=128

> Frame 11805: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF...
> Ethernet II, Src: TPLink_Fd:83:58 (0c:ef:15:fd:83:58), Dst: AzureWaveTec_b6:f1:15 (70:66:55:b6:f1:15)
Destination: AzureWaveTec_b6:f1:15 (70:66:55:b6:f1:15)
...0... = IG bit: Globally unique address (factory default)
...0... = IG bit: Individual address (unicast)
Source: TPLink_Fd:83:58 (0c:ef:15:fd:83:58)
...0... = IG bit: Globally unique address (factory default)
...0... = IG bit: Individual address (unicast)
Type: IPv6 (0x86dd)
[Stream index: 0]
> Internet Protocol Version 6, Src: 2001:4860:4828:7700::, Dst: 2406:b400:3a:b41c:117:22e:ed7a:18f0
> Transmission Control Protocol, Src Port: 443, Dst Port: 19005, Seq: 0, Ack: 1, Len: 0

Specifies if this is an individual (unicast) or group (broadcast/multicast) address (eth.dig), 1 bit

Packets: 24434 - Displayed: 391 (1.6%)

Profile: Default

Hot days ahead 39°C

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4. TCP Analysis Flags

TCP analysis flags were used to identify retransmissions, duplicate acknowledgments, and TCP keep-alive packets. These packets help analyze network reliability and communication issues. Observed Flags: - TCP Retransmission - TCP Duplicate ACK - TCP Keep-Alive

The screenshot displays the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The main window is divided into three panes: Packet List, Packet Details, and Packet Bytes.

Packet List: Shows a list of captured packets. The selected packet is 11757, which is a TCP packet. The list includes columns for No., Time, Source, Destination, Protocol, Length, and Info. The info column shows the packet details, including the source and destination IP addresses, the protocol (TCP), and the flags (RST, Seq=1441, Win=131072, Len=930).

Packet Details: Shows the hierarchical structure of the packet. The selected packet is a TCP packet. The details pane shows the Ethernet II header (Source: TPLink_f8:83:58, Destination: AzureWaveTec_b6:f1:15) and the Internet Protocol Version 6 header (Source: 2404:b800:4003:c01::bc, Destination: 2406:b400:3a:b41c:117:22e:ed7a:18f0).

Packet Bytes: Shows the raw data of the packet in hexadecimal and ASCII. The selected packet is a TCP packet, and the bytes pane shows the raw data in hexadecimal and ASCII.

Conclusion:

Wireshark was successfully used to capture and analyze different types of network traffic. The analysis included HTTP requests, DNS queries, TCP handshake packets, and TCP analysis flags. This task helped in understanding how network protocols communicate and how packet analysis can be used for troubleshooting and security monitoring.